Eya BADRI

Final-Year Electrical Engineering Student – ENIT

Tunis, Tunisia +216 95 334 360 eya.badri@etudiant-enit.utm.tn www.linkedin.com/in/aya-badri

Profile

Final-year Engineering student in **Electrical Engineering** at ENIT (national engineering entrance rank, MP track), currently completing a research-oriented stage ingénieur on AI-based state estimation for Li-ion battery digital twins. Research interests center on **hybrid physics-informed machine learning for embedded physical systems** – combining grey-box modeling, sequence/regression learning, and resource-constrained (TinyML) deployment – applied across battery management, thermal systems, and autonomous robotics. Comfortable owning a problem end-to-end: literature positioning, leakage-free experimental design, statistical validation, and on-target embedded deployment.

Education

National Engineering School of Tunis (ENIT) 2023 – 2027 (Final Year)
Engineering Degree – Electrical Engineering *Tunis, Tunisia*

Preparatory Classes – Mathematics & Physics (MP) 2021 – 2023
National Competitive Entrance Exam to Engineering Schools *Tunisia*

Science Baccalaureate 2021
Mention Très Bien – 16.86/20 *Lycée El Wafa – Cité El Ghazala, Ariana*

Research Experience

ACTIA Engineering Services – Powerlab (LRIIES collaboration, ENIT) Jun. – Aug. 2026
Research Intern – AI Estimation Layer, Hybrid Grey-Box Li-ion Battery Digital Twin *Tunis, Tunisia*

- Identified and corrected a state-reset methodology error in a published stateful-LSTM reference implementation (Pau & Aniballi, STMicroelectronics 2024) through exhaustive bit-exact on-target validation – an independently publishable finding.
- Designed the AI estimation layer (compact FNN, 5→64→32→1) of a hybrid grey-box digital twin, fused with a colleague's Thévenin equivalent-circuit + Extended Kalman Filter physics layer, extending a 2023 IREC baseline (GRU/LSTM/FNN comparison, same lab, same vehicle pack data) as its direct sequel.
- Established a leakage-free chronological train/val/test protocol with multi-seed variance quantification and operational-envelope-aware post-training INT8 quantization calibration – a methodological gap identified across three comparable reference papers in the literature.
- Deployed and validated the estimator on STM32G431RB (CMSIS-NN, INT8) with exhaustive bit-exact hardware-in-the-loop validation across 200,000+ real driving-cycle samples (UDDS, US06), rather than sampling-based checks common in prior work.
- Developed a complementary Ridge-regression State-of-Health estimator and identified a temperature confound and an "energy-as-disguised-capacity" leakage risk in the feature set.
- Co-authoring a manuscript (target submission Aug./Sept. 2026) fusing the AI and physics-based estimation layers.

STEG – Tunisian Company of Electricity and Gas Jul. – Aug. 2025
Engineering Discovery Internship *Tunis, Tunisia*

- Studied energy production/distribution infrastructure, high-voltage substations, and industrial safety standards.

Manuscripts in Preparation

- E. Badri, T. [colleague], et al., *"Estimation de SOC et SOH d'une cellule Li-ion de véhicule électrique avec des algorithmes d'IA et déploiement embarqué"* – hybrid grey-box digital twin fusing AI-based estimation with an EKF physics layer; embedded deployment and hardware-in-the-loop validation. Target submission: Aug./Sept. 2026.

Selected Engineering Projects

Collaborative Target Tracking by Two Drones – Final Year Project 2025 – 2026

- Multi-agent Deep RL system (Q-Learning, Policy Gradient) for trajectory planning, obstacle avoidance and inter-drone coordination (ROS2); hardware deployment planned on Jetson Nano & STM32 under embedded Linux.

Embedded Linux System – BeagleBone Black 2026

- POSIX multi-threaded control loop (timer, GPIO, ADC, PWM) synchronized via pthread conditional variables; Linux kernel driver capturing GPIO pulses via hardware interrupts for real-time signal period measurement (motor-encoder emulation).

Research Keywords

Hybrid/Grey-Box ML:	Physics-informed modeling · Equivalent circuit models · Extended Kalman Filter · Digital twins
Machine Learning:	Sequence models (LSTM/GRU) · FNN · Regression · Time-series · Leakage-free experimental design
Embedded AI:	TinyML/Edge AI · Quantization (INT8) · TFLite Micro · CMSIS-NN · Hardware-in-the-Loop validation
Embedded Systems:	STM32 · BeagleBone Black · Jetson Nano · Kernel drivers · POSIX threads · Real-time systems
Domain Applications:	Li-ion battery management (SOC/SOH) · Autonomous robotics · Multi-agent systems

Technical Skills

Programming	Python, C, C++, Embedded C
ML/Data	TensorFlow/Keras, NumPy, statistical validation, Kaggle
Embedded	STM32CubeIDE, CMSIS-NN, Keil, Jetson Nano, kernel module development
Modeling	MATLAB (ECM/EDA), Extended Kalman Filter, LaTeX
Tools	Git, PuTTY, Eagle CAD, Wokwi, Linux CLI
Certifications	Deep RL – HuggingFace

Activities & Involvement

ENIT Robotics Club 2023 – Present

Sponsorship Manager | Executive Board Member

- Design of a line-following robot; robotics competitions; partnership and sponsorship management.

Aikido – Black Belt 10 years of practice

Martial arts – Discipline, rigor and self-improvement

Languages

Arabic – Native | **French** – B2 | **English** – B2 | **German** – A1